Logistic Regression Report

1. Logistic Regression EDA

The *ICU* dataset contains randomly sampled 200 observations from a larger study on the survival of patients following admission to the Intensive Care Unit (ICU) in a major city. The population of interest are patients who are admitted to an ICU, encompassing both medical and surgical cases, with varying acuity levels from elective admissions to emergency presentations requiring critical care.

Exploratory data analysis revealed the dataset is complete and does not contain any missing values. Moreover, the ICU dataset has minimal outliers in systolic blood pressure (5) and heartrate (1), which could be clinically meaningful for mortality prediction. Patients with extreme vital signs are the highest risk, therefore excluding them would bias the model towards healthier patients. Correlation analysis indicated that level of consciousness (LOC) had the strongest association with survival status with a correlation coefficient of 0.437, suggesting a moderate positive relationship with survival status. Weaker positive associations include type of admission (r = 0.244), CPR needed prior to admission (r = 0.223), age at admission (r = 0.189), and infection suspected prior to admission (r = 0.182).

Importantly, there is the presence of a moderate class imbalance in the response variable STA, with 160 “Lived” observations (80%) and 40 “Died” observations (20%), seen in Figure A1. However, this moderate imbalance is expected in ICU settings, and a standard logistic regression model can be used for the final sample. Thus, no dropout conditions or data manipulation is needed.

2. Contingency Table 1: Gender & Survival Status

A contingency table, Table 1, between gender (SEX) and survival status (STA) was created to determine the odds ratio (OR) of ICU survival based on gender. First, we must calculate the survival odds for each gender category. To calculate odds of males surviving, we find the ratio of the probabilities for successes (i.e., lived) over failures (i.e., died). For males, the odds of survival are 4.167, while for females, the odds of survival are 3.750. To find the OR, we simply divide the odds for males over the odds for females, which equated to 1.111. An OR equal to 1 indicates there no difference between the groups, therefore, an OR of 1.111 indicates slightly higher odds of survival for males over female, indicate very little difference among between gender groups on whether a patient survives.

Table 1. Contingency table of SEX and STA (Survival status)

	Status		
Gender	Lived	Died	**Sum**
Male	100	24	124
Female	60	16	66
Sum	160	40	200

3. Contingency Table 2: Type of Admission & Status

Another contingency table, Table 2, between type of ICU admission (TYP) and survival status (STA) was developed to establish the OR of survival based on ICU admission type. Like the first contingency table, the odds for each admission type were calculated by dividing the probabilities of surviving over dying. The odds of survival for elective ICU admission are 25.500, while the odds of survival for emergency ICU admission are 2.868. The OR for survival rate based on admission type corresponded to 8.890, well over the OR of 1. This is a substantial increase in survival rates for elective ICU admission types compared to emergency ICU admission, indicating a clear effect of survival based on admission type. This increase is naturally intuitive as emergency admissions may need more critical care and have more drastic conditions upon ICU admission as opposed to elected ICU admission.

Table 2. Contingency Table of Admission Type and Survival Status

	Status		
Admission Type	Lived	Died	**Sum**
Elective	51	2	53
Emergency	109	38	147
Sum	160	40	200

4. Logistic Regression

To investigate whether patient age is a meaningful predictor of ICU survival status (STA), a simple logistic regression was fitted with *STA* as the binary response variable (0 = Lived, 1 = Died) and *AGE* as the continuous predictor variable. The logistic regression model is written as:

$$\mathrm{P(STA = 1 \mid AGE)} = e^{\beta_0+\beta_1*AGE}/(1 + \mathrm{e}^{\beta_0+\beta_1*AGE})$$

and the corresponding logit transformation can be written as:

$$\hat{\pi}(AGE) = \widehat{\beta_0} + \widehat{\beta_1} * AGE$$

that represents the log-odds of death as a linear function of age.

4.1 AGE Variable Analysis

A scatterplot of STA by AGE was produced to visually assess whether age discriminates between those who lived and those who died seen in Figure A2. Because STA is binary, all data points fall on either Y = 0 or Y = 1, making direct visual interpretation difficult. However, tendency for deaths (red points) to cluster at older ages, suggesting age may have some discriminating power, though the pattern is not immediately striking from the raw scatterplot alone. To better understand the relationship, AGE was discretized into nine decade-based categories (AGE_CAT 1 through 9, spanning ages 15–24 up to 95 and over), and the mean STA (i.e., proportion of deaths) was computed within each group. The resulting bar chart, Figure A3, reveals a clear and increasing trend in mortality proportion across age groups, with the youngest patients showing near-zero mortality and the oldest groups, particularly those aged 75 and above, showing substantially higher mortality proportions. This confirms that the relationship between

age and death, while not perfectly linear, is broadly systemically increasing, supporting the use of AGE as a predictor.

4.2 Model Interpretation

The fitted logistic regression model produces estimated coefficients of approximately -3.06 for the intercept and 0.028 for AGE. The intercept, while the intercept was significant ($z = -4.394$, $p < 0.0001$), represents the log-odds of death when AGE is zero, no practical meaning but necessary for the model. The slope coefficient of 0.028 indicates that for each one-year increase in age, the log-odds of death increase by approximately 0.028, corresponding to an odds ratio of $e^{0.028} \approx 1.028$. Hypothesis testing for the AGE coefficient yielded a statistically significant result ($p < 0.0001$) based on both the Wald z-test from the model summary and the likelihood ratio test from the ANOVA output, allowing us to reject the null hypothesis that $\beta_1=0$. We conclude that there is strong statistical evidence that age is a significant predictor of ICU mortality.

The model's AIC and BIC values were approximately 200 and 207, respectively, and the residual deviance was approximately 196, compared to a null deviance of approximately 200. While the deviance reduction confirms that adding AGE improves model fit relative to an intercept-only model, the modest magnitude of this reduction suggests that AGE alone does not explain a large share of the variability in patient outcomes.

The predicted logit values were plot against AGE, seen in Figure A4, produce a perfectly linear function and form a straight line, which is expected with the logit equation. Additionally, the predicted probabilities were plotted against AGE with the raw STA values overlaid in red, shown in Figure A5. The resulting curve does not display the full classic S-shape associated with logistic regression, because the predicted probabilities across the age range of the dataset only span roughly 0.10 to 0.46, suggesting that only the lower-left portion of the logistic curve is visible. The increase in predicted mortality appears gradual at younger ages and becomes steeper after approximately age 75, which is consistent with the pattern seen in the grouped bar chart.

4.3 Probability of Survival

The fitted model was used to predict the probability of ICU death for a 29-year-old patient. The predicted probability was approximately 0.095, meaning the model assigns roughly a 9.5% chance of death for a patient of that age. This prediction is consistent with what is seen in the scatterplot, where the fitted curve at age 29 lies quite low. Notably, the 25–34 age group had zero observed deaths in the sample, which might suggest an even lower true probability; however, the logistic model appropriately smooths across all ages rather than assigning a probability of exactly zero, which would be statistically inappropriate given the small sample size in that group.

This prediction and the model overall should be interpreted with caution. A 9.5% predicted mortality for a young adult ICU patient may seem reasonable in isolation, but age alone is a limited predictor of such a complex clinical outcome. ICU mortality is influenced by a wide range of factors including diagnosis, vital signs, comorbidities, and prior health status, none of which are captured in this univariate model. This simple model serves as a useful starting point,

but additional covariates would almost certainly be needed to produce a model with clinically meaningful predictive accuracy.

5. Reflection & Next Steps

The natural next step is to expand the model by incorporating additional clinical and demographic variables available in the ICU dataset. The modest deviance reduction observed in the univariate model makes clear that age alone captures only a small fraction of the variability in patient survival, and a more comprehensive model is needed to improve predictive accuracy and clinical relevance. The next phase begins with exploratory data analysis of the remaining candidate predictors in the dataset. This means examining each variable individually, both continuous variables like vital signs and categorical variables like diagnosis codes, gender, and admission type, in relation to STA, much as we did with AGE. This univariate screening phase helps identify which variables are worth carrying forward into a multivariable model and whether any transformations or re-codings are needed before doing so.

Once candidate predictors have been identified and understood individually, the next step is to build a multivariable logistic regression model. It is important to assess potential interactions between predictors as well, since the effect of age on mortality may differ depending on, for example, a patient's diagnosis or whether they were conscious on admission. After a multivariable model has been fitted, formal model diagnostics and goodness-of-fit assessment become critical. This includes examining influence diagnostics such as Cook's distance and leverage values to identify observations that exert undue influence on the model and checking whether the linearity assumption holds for continuous predictors on the logit scale.

A. Appendix

Figure 1. Survival Status Variable Distribution

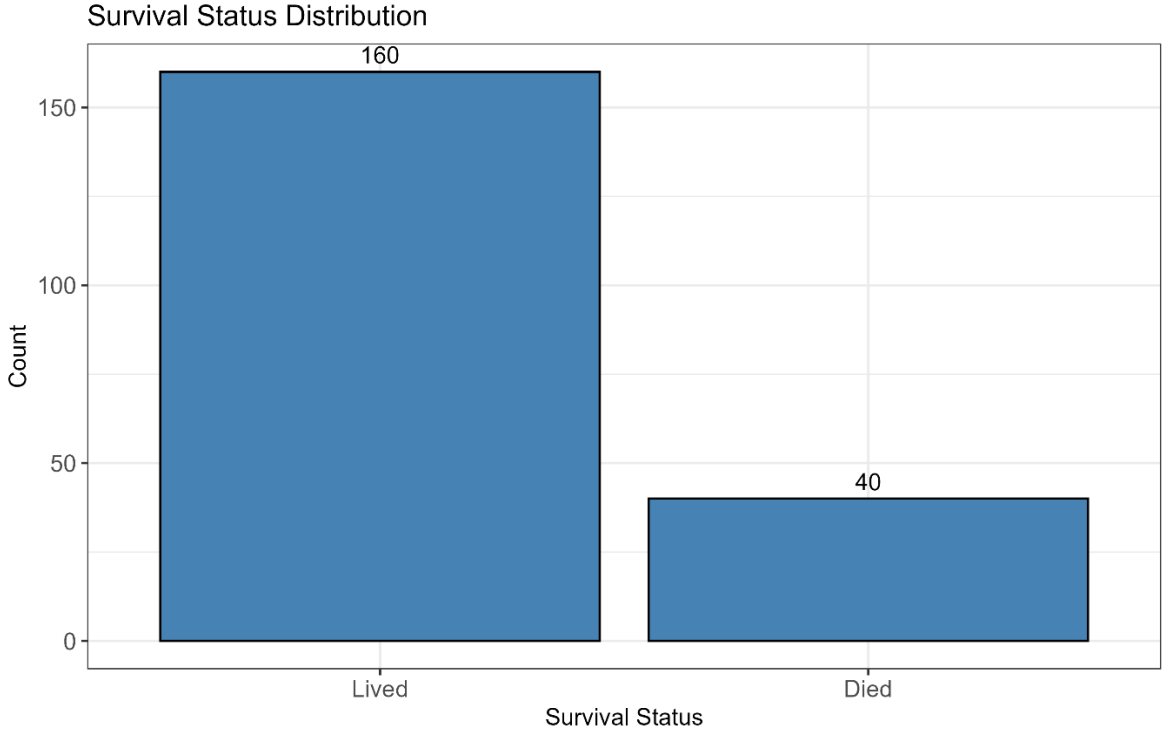

Figure 2. Scatterplot of Age and Survival Status

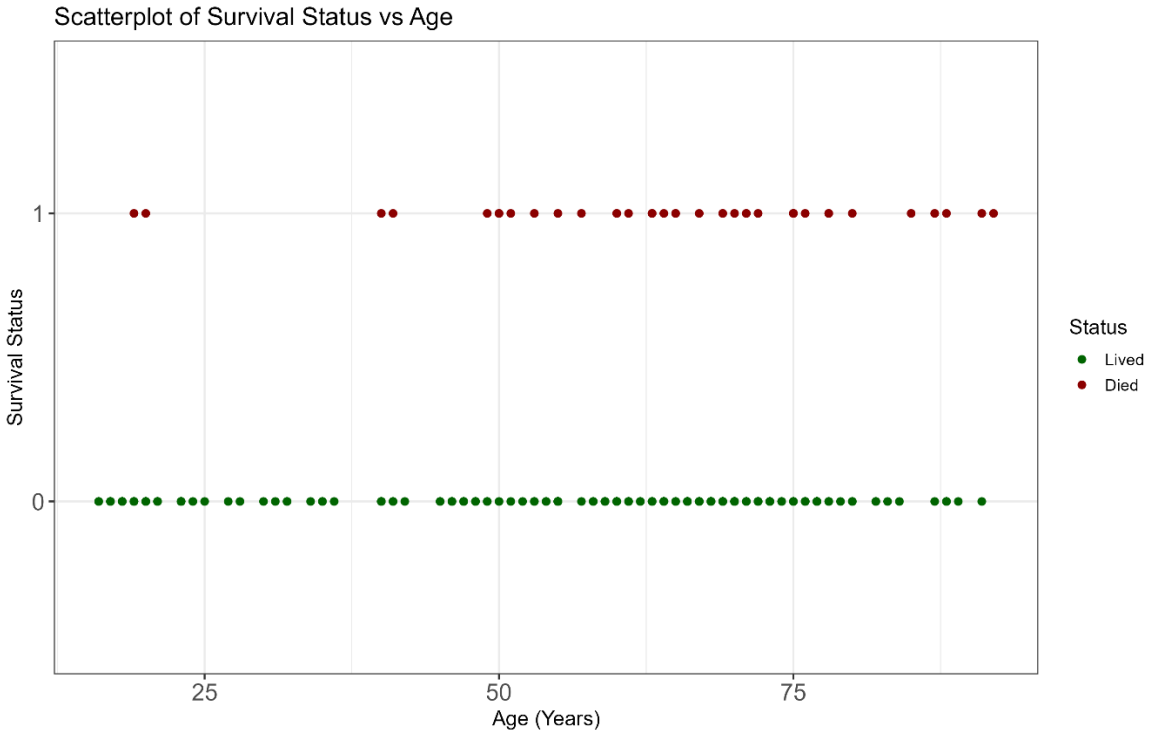

Figure 3. Grouped Bar of Mean STA by Age Category

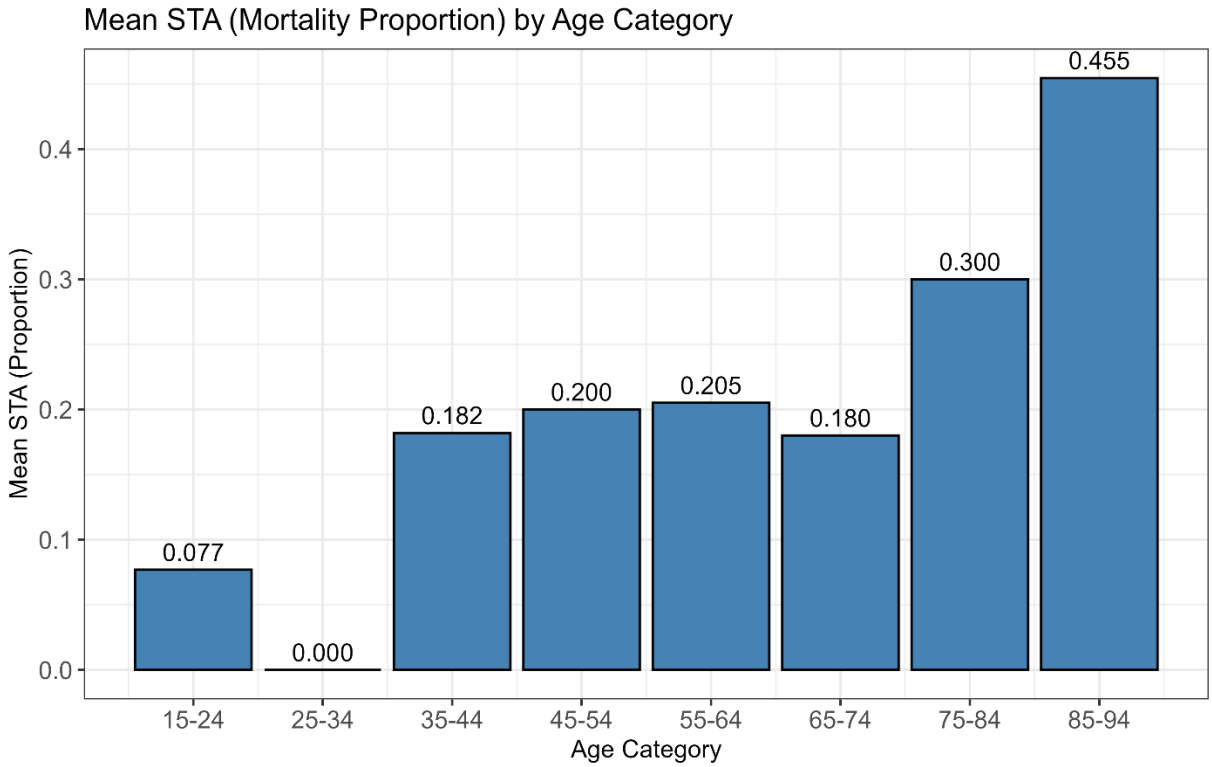

Figure 4. Scatterplot of Predicted Values Versus Age

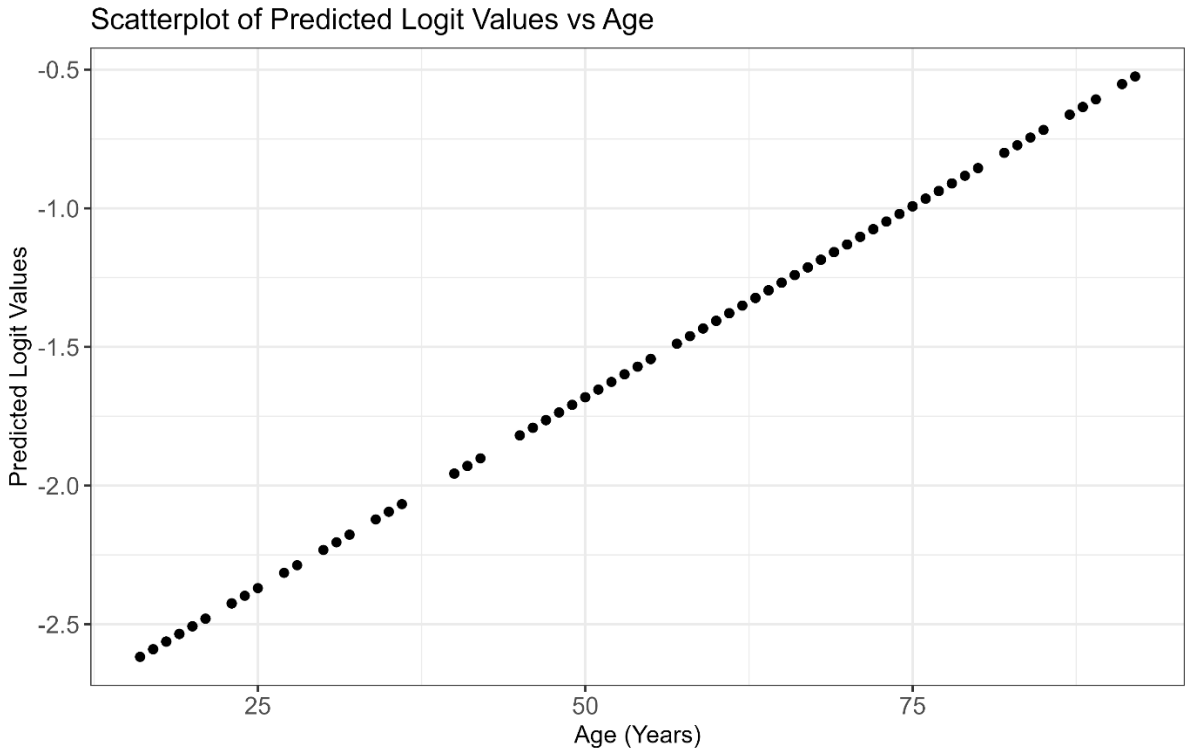
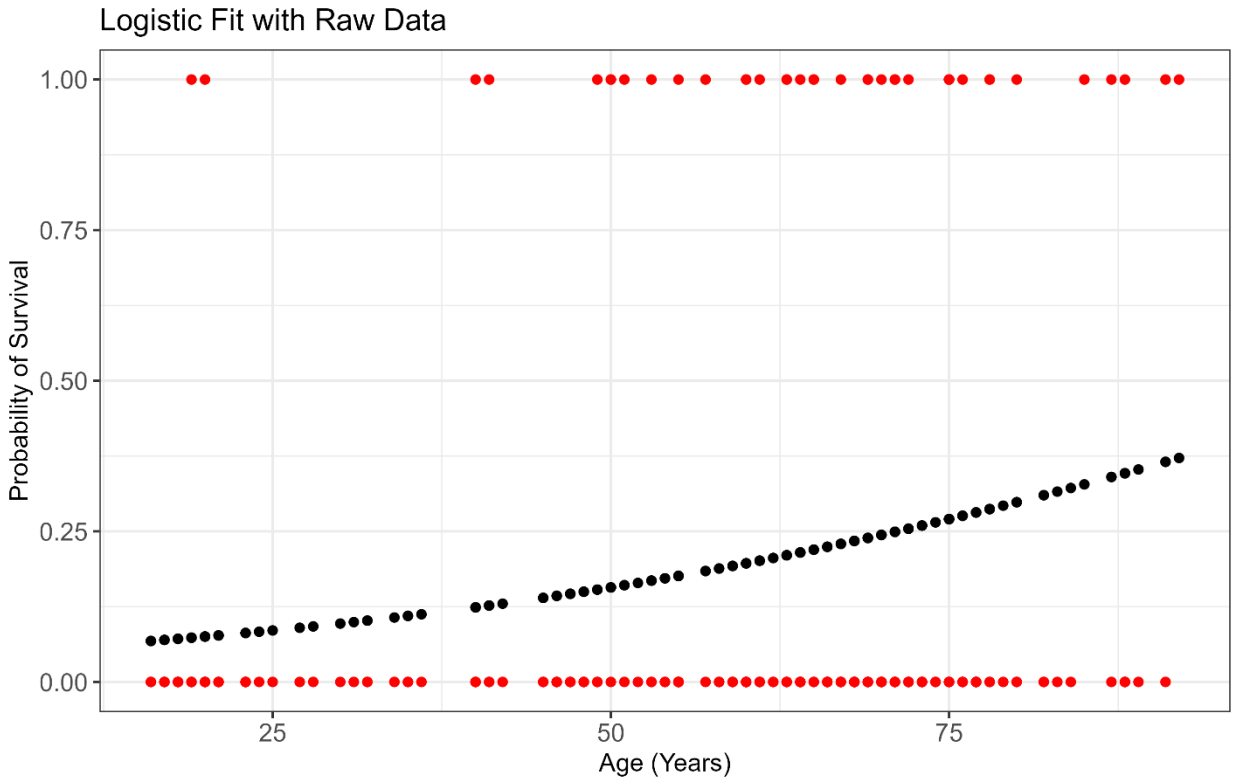
Figure 5. Logit Probabilities with Original STA Values Versus Age